



Lesson 6.8 — Solving Logarithmic Equations

Big idea: Combine logs, then exponentiate. ALWAYS check answers — log domains exclude non-positive arguments, and combining logs can introduce extraneous solutions.

Quick review

Use the rules (product/quotient/power) to combine multiple logs into a single log.

Then exponentiate: $\log_b(\text{stuff}) = c \Rightarrow \text{stuff} = b^c$.

Check domain: every original log argument must be positive.

One-to-one: $\log_b(M) = \log_b(N) \Rightarrow M = N$ (provided both arguments are positive).

Worked example

Worked example. Solve $\log(x) + \log(x - 3) = 1$.

$$\log[x(x - 3)] = 1 \Rightarrow x(x - 3) = 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x - 5)(x + 2) = 0. \quad x = 5 \text{ (reject } -2: \log \text{ undefined).}$$

Warm-ups

1. Solve $\log_2(x) = 4$.
2. Solve $\log(x) = 2$.
3. Solve $\ln(x) = -1$.

Practice

1. Solve $\log_3(x - 2) = 2$.
2. Solve $\log(x) + \log(x - 9) = 1$.
3. Solve $2 \ln(x) - \ln(x - 3) = 0$.

Challenge

1. Solve $\log_2(x + 1) - \log_2(x - 1) = 2$.
2. Solve $\log_2(x) \cdot \log_x(8) = 6$ (be careful with the change of base trick).



Answer key — Lesson 6.8

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x = 16$.
2. $x = 100$.
3. $x = e^{-1} = 1/e \approx 0.368$.

Practice

1. $x - 2 = 9 \Rightarrow x = 11$.
2. $\log[x(x - 9)] = 1 \Rightarrow x(x - 9) = 10 \Rightarrow x^2 - 9x - 10 = 0 \Rightarrow (x - 10)(x + 1) = 0 \Rightarrow x = 10$ (reject -1).
3. $\ln(x^2/(x - 3)) = 0 \Rightarrow x^2/(x - 3) = 1 \Rightarrow x^2 = x - 3 \Rightarrow x^2 - x + 3 = 0$. Discriminant $= 1 - 12 < 0 \rightarrow$ no real solution.

Challenge

1. $\log_2[(x + 1)/(x - 1)] = 2 \Rightarrow (x + 1)/(x - 1) = 4 \Rightarrow x + 1 = 4x - 4 \Rightarrow x = 5/3$. Check $x > 1$: ✓.
2. $\log_x(8) = \ln 8 / \ln x = 3 \ln 2 / \ln x$. $\log_2(x) = \ln x / \ln 2$. Product $= 3 \ln 2 / \ln 2 = 3$? That equals 3, not 6. The product $\log_2(x) \cdot \log_x(8)$ actually simplifies to $\log_2(8) = 3$ (chain rule for logs). So the equation becomes $3 = 6$, which is false — NO solution. (The equation has no solution for $x > 0$, $x \neq 1$.)